CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 26: Co-ordinate Geometry

Topic-wise short notes with solved examples

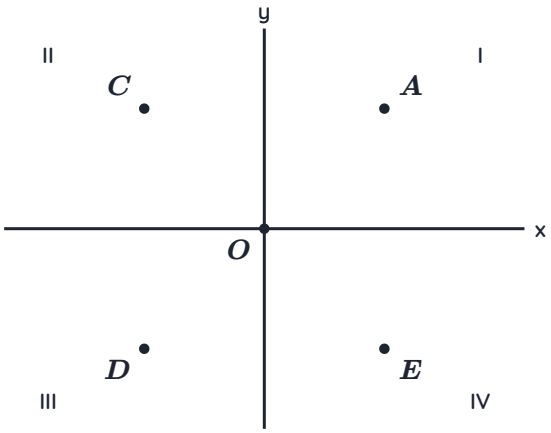
The Cartesian plane, co-ordinates of points, graphs of linear equations, and the slope and intercept of a straight line.

1 The Cartesian plane

- Two perpendicular number lines, the **x-axis** (horizontal) and the **y-axis** (vertical), meet at the **origin** $O(0, 0)$.
- The position of a point is given by an **ordered pair** (x, y). The first number is the **abscissa** (x co-ordinate): its distance from the y-axis. The second is the **ordinate** (y co-ordinate): its distance from the x-axis.
- The order matters: $(2, 5)$ and $(5, 2)$ are different points.
- The axes divide the plane into four **quadrants**:

Quadrant	Sign of x	Sign of y	Example
I	$+$	$+$	$(3, 4)$
II	$-$	$+$	$(-3, 4)$
III	$-$	$-$	$(-3, -4)$
IV	$+$	$-$	$(3, -4)$

- A point on the **x-axis** has $y = 0$: $(a, 0)$. A point on the **y-axis** has $x = 0$: $(0, b)$.



A(3, 3), C(−3, 3), D(−3, −3), E(3, −3) lie in quadrants I, II, III, IV

Example 1.1 In which quadrant or on which axis do these points lie: $(-2, 5)$, $(4, -1)$, $(0, -3)$, $(-6, 0)$?

SOLUTION

Quadrant II; quadrant IV; on the y-axis; on the x-axis

Example 1.2 Find x and y if $(2x - 1, y + 3) = (5, 7)$.

SOLUTION

Equal ordered pairs have equal co-ordinates: $2x - 1 = 5$ and $y + 3 = 7$.

$x = 3, y = 4$

Example 1.3 A(2, 1), B(7, 1) and C(7, 4) are three vertices of a rectangle ABCD. Find D and the area.

SOLUTION

D has the x co-ordinate of A and the y co-ordinate of C. $AB = 5$ and $BC = 3$.

$D(2, 4)$; **area 15 square units**

2 Dependent and independent variables

- In $y = 3x + 2$, the value of y depends on the value chosen for x. Here x is the **independent variable** and y the **dependent variable**.
- If the equation is solved for x, as in $x = \frac{y - 2}{3}$, the roles are reversed.

3 Graph of a linear equation

- The graph of $ax + by + c = 0$ is a **straight line**. Every point on the line satisfies the equation, and every solution lies on the line.
- **To draw it**: make y the subject; choose three convenient values of x and find y; plot the three points; join them with a ruler. Two points fix the line; the third is a check.

Equation	Graph
$x = 0$	The y-axis
$y = 0$	The x-axis
$x = a$	A line parallel to the y-axis, a units from it
$y = b$	A line parallel to the x-axis, b units from it
$y = mx$	A line through the origin

Example 3.1 Make a table of three points for the graph of $2x - y = 4$.

SOLUTION

$y = 2x - 4$. For $x = 0, 2, 3$: $y = -4, 0, 2$.

$(0, -4), (2, 0), (3, 2)$

Example 3.2 Does the point $(3, -1)$ lie on the line $2x + 3y = 3$?

SOLUTION

$2(3) + 3(-1) = 3$

Yes

Example 3.3 The point $(k, 4)$ lies on the line $3x - 2y = 1$. Find k.

SOLUTION

$3k - 8 = 1$

$k = 3$

Example 3.4 Find the area of the triangle formed by the line $3x + 4y = 12$ and the axes.

SOLUTION

The line meets the x-axis at $(4, 0)$ and the y-axis at $(0, 3)$.

$\frac{1}{2} \times 4 \times 3 = 6$ **square units**

4 Inclination and slope

- The **inclination** θ of a line is the angle it makes with the **positive direction of the x-axis**, measured anticlockwise.
- The **slope** (gradient) is $m = \tan\theta$.

Inclination	Slope
$0°$ (parallel to the x-axis)	0
$30°$	$\frac{1}{\sqrt{3}}$
$45°$	1
$60°$	$\sqrt{3}$
$90°$ (parallel to the y-axis)	Not defined

- A line rising to the right has a positive slope. A line falling to the right has a negative slope.

Example 4.1 Find the slope of a line whose inclination is $60°$, and the inclination of a line whose slope is 1.

SOLUTION

$\sqrt{3}$; $45°$

5 The form y = mx + c

- When the equation of a line is written as $y = mx + c$:
- m is the **slope**.
- c is the **y-intercept**: the line cuts the y-axis at $(0, c)$.
- To find the slope and the y-intercept of any line, make y the subject.
- The **x-intercept** is found by putting $y = 0$.
- Lines with the **same slope** are parallel.

Example 5.1 Find the slope and the y-intercept of $3x + 4y = 8$.

SOLUTION

$y = -\frac{3}{4}x + 2$

Slope $-\frac{3}{4}$; y-intercept 2

Example 5.2 Write the equation of the line with slope 2 and y-intercept -5.

SOLUTION

$y = 2x - 5$

Example 5.3 Find the equation of the line with inclination $45°$ that cuts the y-axis at $(0, 3)$.

SOLUTION

$m = \tan 45° = 1$ and $c = 3$

$y = x + 3$

Example 5.4 Are the lines $2x - y = 5$ and $4x - 2y = 3$ parallel?

SOLUTION

$y = 2x - 5$ and $y = 2x - \frac{3}{2}$: both have slope 2.

Yes

Example 5.5 The line $y = mx + c$ passes through $(0, 4)$ and $(2, 10)$. Find m and c.

SOLUTION

From $(0, 4)$: $c = 4$. From $(2, 10)$: $10 = 2m + 4$.

$m = 3$, $c = 4$

6 Common mistakes

- Plotting (x, y) with the two numbers interchanged.
- Saying a point such as $(0, 5)$ lies on the x-axis. It is on the y-axis, since $x = 0$.
- Reading the slope from $ax + by + c = 0$ as a. Make y the subject first: the slope is $-\frac{a}{b}$.
- Taking the sign of the intercept wrongly: in $y = 2x - 5$ the y-intercept is -5.
- Drawing $x = 3$ parallel to the x-axis. It is parallel to the y-axis.
- Joining plotted points freehand instead of with a ruler.

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